

ALFAA31244

## Di-n-butyl phthalate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 邻苯二甲酸二正丁酯  
**Di-n-butyl phthalate**

**Cat No. :** 31244  
**Synonyms** DBP; n-Butyl phthalate  
**CAS No** 84-74-2  
**Molecular Formula** C16 H22 O4

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against**

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
Odorless

#### Emergency Overview

May damage fertility or the unborn child. Very toxic to aquatic life.

#### Classification of the substance or mixture

|                        |             |
|------------------------|-------------|
| Reproductive Toxicity  | Category 1B |
| Acute aquatic toxicity | Category 1  |

#### Label Elements



**Signal Word**

**Danger**

**Hazard Statements**

H360 - May damage fertility or the unborn child  
H400 - Very toxic to aquatic life

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May damage fertility or the unborn child.

**Environmental hazards**

Very toxic to aquatic life. . Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.  
The product is insoluble and sinks in water.

**Other Hazards**

Contains a known or suspected endocrine disruptor. Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties. Contains a substance on the National Authorities Endocrine Disruptor Lists. Toxic to terrestrial vertebrates.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| Dibutyl phthalate | 84-74-2 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of

## Di-n-butyl phthalate

contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep container tightly closed. Keep in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component         | China                      | Taiwan                   | Thailand | Hong Kong |
|-------------------|----------------------------|--------------------------|----------|-----------|
| Dibutyl phthalate | TWA: 2.5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> |          | -         |

| Component         | ACGIH TLV                | OSHA PEL         | NIOSH                        | The United Kingdom            | European Union |
|-------------------|--------------------------|------------------|------------------------------|-------------------------------|----------------|
| Dibutyl phthalate | TWA: 5 mg/m <sup>3</sup> | (Vacated) TWA: 5 | IDLH: 4000 mg/m <sup>3</sup> | STEL: 10 mg/m <sup>3</sup> 15 |                |

**SAFETY DATA SHEET****Di-n-butyl phthalate**

|  |  |                                               |                          |                                      |  |
|--|--|-----------------------------------------------|--------------------------|--------------------------------------|--|
|  |  | mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> | min<br>TWA: 5 mg/m <sup>3</sup> 8 hr |  |
|--|--|-----------------------------------------------|--------------------------|--------------------------------------|--|

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection** Goggles (European standard - EN 166)**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments                           |
|----------------|-------------------|-----------------|-------------|------------------------------------------|
| Nitrile rubber | > 480 minutes     | 0.1 - 0.2 mm    | Level 6     | As tested under EN374-3 Determination of |
| Neoprene       | > 480 minutes     | 0.45 mm         | EN 374      | Resistance to Permeation by Chemicals    |
| Butyl rubber   | > 480 minutes     | 0.35 mm         |             |                                          |
| Viton (R)      | > 480 minutes     | 0.30 mm         |             |                                          |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

## Di-n-butyl phthalate

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | Not applicable           |                                          |
| <b>Melting Point/Range</b>                     | -35 °C / -31 °F          |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 340 °C / 644 °F          |                                          |
| <b>Flash Point</b>                             | 157 °C / 314.6 °F        | <b>Method</b> - No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.043                    |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | 0.13 g/l (20°C)          | practically insoluble                    |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Dibutyl phthalate                              | 4.79                     |                                          |
| <b>Autoignition Temperature</b>                | 390 °C / 734 °F          |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |
| <b>Molecular Formula</b>                       | C16 H22 O4               |                                          |
| <b>Molecular Weight</b>                        | 278.34                   |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.          |
| <b>Hazardous Reactions</b>      | None under normal processing.            |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Conditions to Avoid</b>      | Incompatible products.                   |
| <b>Materials to avoid</b>       | Acids. Bases. Chlorine.                  |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component         | LD50 Oral                 | LD50 Dermal             | LC50 Inhalation                |
|-------------------|---------------------------|-------------------------|--------------------------------|
| Dibutyl phthalate | LD50 = 7499 mg/kg ( Rat ) | >20000 mg/kg ( Rabbit ) | LC50 >= 15.68 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;****Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

**(g) reproductive toxicity;****Reproductive Effects**

Category 1B Based on available data, the classification criteria are not met

**Developmental Effects**

Possible risk of impaired fertility.

**Teratogenicity**

Component substance is listed on California Proposition 65 as a developmental hazard.

May cause harm to the unborn child.

**(h) STOT-single exposure;**

Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;**

Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** No information available**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

| Component         | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                 | Water Flea                                                                             | Freshwater Algae                                                                                                  | Microtox                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Dibutyl phthalate | LC50: 0.42 - 1.28 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 1.24 - 5.3 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: > 1.24 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.31 - 5.45 mg/L, 96h static (Pimephales promelas)<br>LC50: 0.71 - 1.2 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 1.38 - 1.74 mg/L, 96h flow-through (Lepomis macrochirus) | EC50: = 3.4 mg/L, 48h (Daphnia magna)<br>EC50: = 2.99 mg/L, 48h Static (Daphnia magna) | EC50: = 0.4 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: = 1.2 mg/L, 72h (Desmodesmus subspicatus) | EC50 = 10.9 mg/L 30 min<br>EC50 = 10.9 mg/L 5 min<br>EC50 = 11.1 mg/L 15 min<br>EC50 = 2.2 mg/L 24 h |

**Persistence and Degradability****Persistence**

Expected to be biodegradable

**Degradation in sewage treatment plant**

May persist.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**SAFETY DATA SHEET****Di-n-butyl phthalate****Bioaccumulative Potential** Product has a high potential to bioconcentrate

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Dibutyl phthalate | 4.79    | No data available             |

**Mobility in soil** Spillage unlikely to penetrate soil The product is insoluble and sinks in water Is not likely mobile in the environment due its low water solubility Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

**Endocrine Disruptor Information**

| Component         | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Dibutyl phthalate | Group I Chemical                         | High Exposure Concern                            |                                         |

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS**

**Waste from Residues/Unused Products** Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Dibutyl phthalate  
**Hazard Class** 9  
**Packing Group** III

**IMDG/IMO**

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Dibutyl phthalate  
**Hazard Class** 9  
**Packing Group** III

**IATA**

**UN-No** UN3082  
**Proper Shipping Name** Environmentally hazardous substances, liquid, n.o.s.  
**Technical Shipping Name** Dibutyl phthalate  
**Hazard Class** 9  
**Packing Group** III

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION**

## Di-n-butyl phthalate

## International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Dibutyl phthalate | -                                                   | -                                       | X    | X     | 201-557-4 | X    | X   | X     | X    | X    | X    | KE-02214 |

## National Regulations

| Component                             | Toxic Chemical Substances Control Act                |
|---------------------------------------|------------------------------------------------------|
| Dibutyl phthalate<br>84-74-2 ( ≤100 ) | Class I (10 wt%)<br>Class II (10 wt%)<br>TRQ = 50 kg |

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 03-Dec-2010  
**Revision Date** 26-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

## Training Advice

Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**